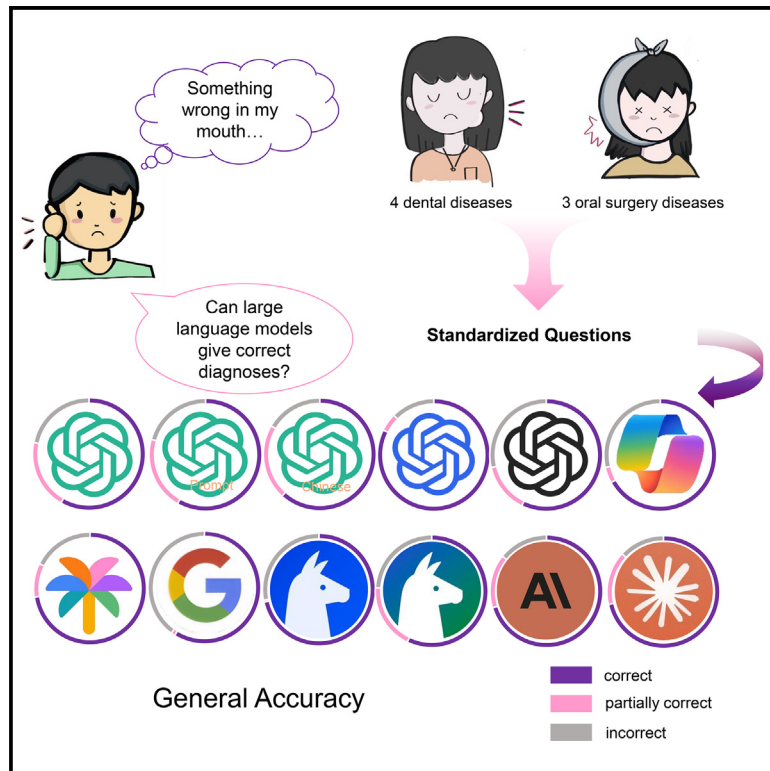


Evaluation of the ability of large language models to self-diagnose oral diseases

Graphical abstract



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In brief

Artificial intelligence; Computing methodology; Health sciences

Highlights

- All LLMs can diagnose TMJ disorders, periodontal disease, dental caries, and malocclusion
- The use of prompts did not significantly affect the diagnostic accuracy of LLMs
- The language differences should be considered when applying LLMs to the self-diagnosis
- All LLMs provided recommendations for seeking medical attention



Article

Evaluation of the ability of large language models to self-diagnose oral diseases

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<https://doi.org/10.1016/j.isci.2024.111495>

SUMMARY

Large language models (LLMs) offer potential in primary dental care. We conducted an evaluation of LLMs' diagnostic capabilities across various oral diseases and contexts. All LLMs showed diagnostic capabilities for temporomandibular joint disorders, periodontal disease, dental caries, and malocclusion. The prompts did not affect the performance of ChatGPT 3.5. When Chinese was used, the diagnostic ability of ChatGPT 3.5 for pulpitis improved (0% vs. 61.7%, $p < 0.001$), while the ability to diagnose pericoronitis decreased (8% vs. 0%, $p < 0.001$). For ChatGPT 4.0 in Chinese, they were both improved (0% vs. 92%, 8% vs. 72%, $p < 0.001$, respectively). Claude 2 exhibited the highest accuracy in diagnosing pulpitis (36%, $p = 0.048$), ChatGPT 4.0 showed complete diagnostic capability for pericoronitis. Llama 2 and Claude 3.5 Sonnet exhibited complete diagnostic capability for oral cancer. In conclusion, LLMs may be a potential tool for daily dental care but need further updates.

INTRODUCTION

According to the World Health Organization's Global Oral Health Status Report,¹ nearly half of the global population (approximately 3.5 billion people) experiences varying degrees of oral health issues. However, approximately 75% of these individuals reside in low- or middle-income countries. Over the past 30 years, the number of cases of oral diseases worldwide has increased by one billion, clearly indicating that many individuals lack access to preventive and treatment services for oral health issues. Among all oral diseases, the most prevalent include dental caries, severe periodontal diseases, tooth loss, and oral cancer. Untreated dental caries, affecting approximately 2.5 billion people, is the most common global ailment among oral diseases. Severe periodontal diseases are a primary cause of tooth loss and are estimated to impact approximately one billion people globally. Additionally, each year, approximately 380,000 patients are newly diagnosed with oral cancer.

Effective oral diagnostics and medical advice are crucial for the promotion of oral health services, especially for populations

in developing countries. However, many elderly individuals² and people in developing countries³ lack awareness of oral care. The provision of oral health services often depends heavily on highly specialized providers using expensive high-tech equipment and materials, which are not well integrated into primary health care models. Currently, there is an urgent need for a low-cost, simple, effective, as well as personalized method for oral diagnostics that can be applied in daily life.^{4,5}

In recent years, artificial intelligence (AI) has undergone significant development, and its applications in medicine have garnered widespread attention. Among these applications, the use of large language models (LLMs) has become particularly convenient because it allows for AI interaction through natural language conversations. In the decade leading up to the significant attention garnered by LLMs, the field of language modeling has undergone numerous revolutionary changes. Initial natural language models were based on n-gram modeling, a probabilistic yet effective approach, particularly in medical research. However, these models were confined to medium scales due to constraints in architecture design and hardware support, despite the proposal



of efficient algorithms for the medical domain. Among the most notable LLMs is Google's T5, which boasts 11 billion parameters. Its competitor, ChatGPT, developed by OpenAI, is even larger, with 175 billion parameters. These billion-parameter models have ushered in a new era for LLMs and their potential applications.⁶ Although ChatGPT was not explicitly designed for health care, it is increasingly being utilized in health care environments. In the medical field, researchers have employed LLMs such as the generative pretrained transformer chatbot ChatGPT (OpenAI, USA) to assist in extracting unstructured information from medical records for patient classification⁷ or to generate medical-related correspondence.⁸

As a result, there is potential for leveraging LLMs for simple self-diagnosis in oral health, particularly in primary dental care.⁹ Currently, the application of LLMs is mostly focused on oral and maxillofacial surgery and is mainly limited to drafting, education, and communication.¹⁰ There are few studies discussing the use of LLMs in common oral diseases, including periodontology, endodontology, and orthodontics. Additionally, dental diagnosis is a blank space. This situation presents two conflicting challenges. First, there is a current lack of research focused on the application of LLMs in medical diagnostics, especially in oral health diagnostics, leading to concerns within both society and the medical community regarding the disease classification and diagnostic capabilities of LLMs.¹¹ Second, developers of LLMs lack reports assessing the medical capabilities of these models, making it challenging to make targeted improvements. To bridge this gap, an open-domain oral disease Q&A dataset is needed. The dataset should include the full conversation with LLMs and human annotations in standardized format.

Due to the constant evolution of information, LLMs must be re-trained based on the latest data, leading to the emergence of multiple iterative versions. Taking ChatGPT as an example, there are currently multiple versions, such as 3.5 and 4o. While ChatGPT is the most well-known LLM, there are other notable LLMs worth considering. For instance, Copilot (Microsoft, USA) and Bard (Google, USA) can search for answers online, while other LLMs could not.

Moreover, as globally oriented AI products, it remains unclear whether the performance of LLMs in languages other than English is comparable. However, most studies have only researched the performance of LLMs in education or examination in multi languages.^{12–14} Research on the clinical application of LLMs in non-English environments is still rare¹⁵ and warrants further investigation.

LLMs exhibit significant potential for enhancing primary dental care. However, their application in common oral diseases remains underexplored,¹⁶ with systematic comparisons among diverse LLMs lacking. To address this gap and evaluate the feasibility of LLMs as tools for daily self-dental care, we conducted a comprehensive study assessing the capacity of various LLMs to self-diagnose common oral diseases across multiple contexts. This investigation aims to elucidate the efficacy and limitations of LLMs in oral health self-assessment, potentially informing future developments in accessible dental care technologies.

RESULTS

Diagnostic accuracy of ChatGPT 3.5

In the 5-day experiment, ChatGPT 3.5 yielded 102 (58.3%) correct diagnoses and 35 (20%) partially correct diagnoses. ChatGPT 3.5 exhibited a complete inability to diagnose pulpitis (0/25, 0%) (Figure 1A) and preliminary diagnostic capabilities for temporomandibular joint (TMJ) disorders, periodontal disease, caries, and malocclusion (25/25, 100%) (Figures 1B–1F). For pericoronitis, ChatGPT 3.5 provided 14 (64%) partially correct diagnoses. However, there was a tendency for ChatGPT 3.5 to describe symptoms related to pericoronitis as periodontal disease and prioritize periodontal disease as the primary diagnosis. For oral cancer, ChatGPT 3.5 only failed to provide a correct diagnosis twice, but its diagnoses were conservative, with oral cancer being considered as the primary diagnosis only 2 times (8%) and as a secondary diagnosis 21 times (84%).

In terms of medical recommendations, ChatGPT 3.5 recommended seeking dental care for all diseases (Tables S1–S7), with the majority describing seeking medical attention as “important,” “crucial,” or “essential.” Notably, the recommendations of ChatGPT 3.5 for malocclusion were relatively mild, including terms such as “can consider,” “appropriate,” and “advisable.” In contrast, recommendations for seeking a dentist for oral cancer were more assertive, including terms such as “particularly important,” “strongly advisable,” and “strongly recommended.”

Differences in diagnostic accuracy among multiple LLMs

To assess the diagnostic capabilities of other LLMs for oral diseases, we selected five other popular LLMs for comparison with ChatGPT 3.5. Table 1 shows the overall accuracy of these LLMs. The experimental results indicate that all selected LLMs can provide preliminary diagnoses for TMJ disorders, periodontal disease, caries, and malocclusion (Figures S1–S9). In the case of pulpitis, the diagnostic ability of Claude 2 was greater than that of ChatGPT 3.5 ($p = 0.04$), providing 9 partially correct diagnoses (36%). (Figure 2A) Regarding symptoms suggestive of pericoronitis, ChatGPT 4.0 ($p < 0.001$) and Claude 2 ($p = 0.02$) exhibited significantly improved diagnostic abilities compared to ChatGPT 3.5. Specifically, Claude 2 achieved 10 correct diagnoses (40%) and 14 partially correct diagnoses (56%) within the 5-day period, while ChatGPT 4.0 showed a comprehensive ability to provide preliminary diagnoses for pericoronitis (25/25, 100%). PaLM 2 showed 12 correct diagnoses (48%) and 6 partially correct diagnoses (24%) but did not show statistical differences with ChatGPT 3.5.

For oral cancer diagnosis, each LLM performed variously and all showed statistical differences. Llama 2 could completely give correct diagnosis (25/25, 100%) ($p < 0.001$). For ChatGPT 4.0, it provided 19 correct diagnoses (76%) and 6 partially correct diagnoses (24%), $p = 0.008$. For PaLM 2, it was 15(60%) and 9(36%), with $p = 0.03$; for Claude 2, it was 18(72%) and 4(16%), with $p = 0.008$; for Copilot, it was 17(68%) and 7(28%), with $p = 0.02$, respectively (Figure 2C). All LLMs recommended seeking a dentist (Tables S22–S56).

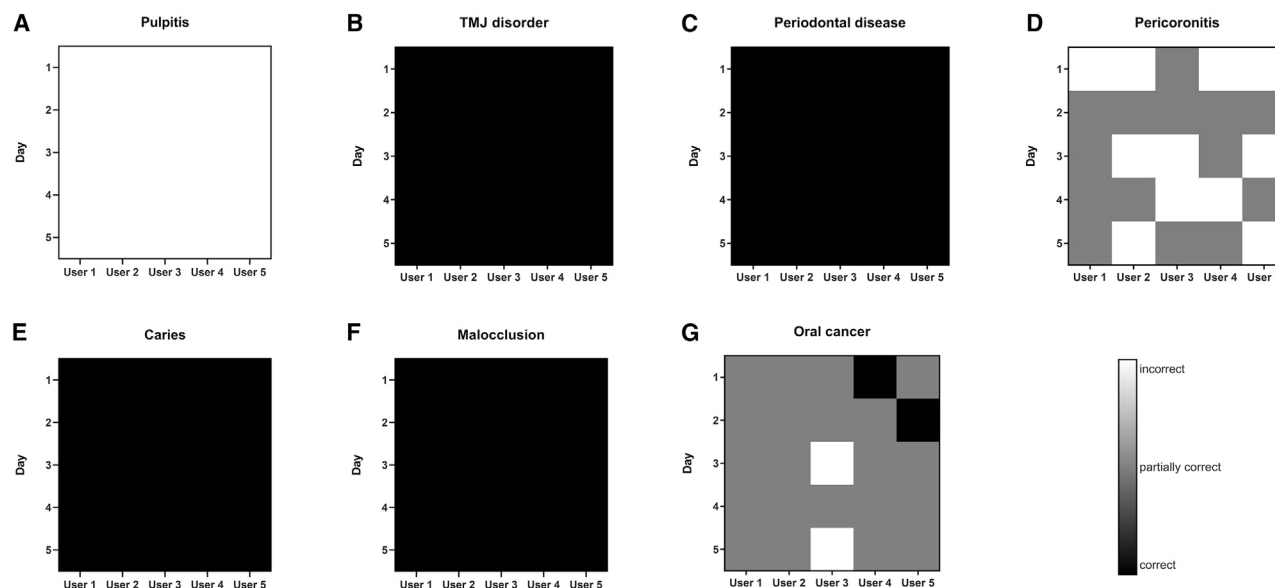


Figure 1. The accuracy of ChatGPT 3.5 in diagnosing common oral diseases

In a five-day experiment, five participants individually inputted formulated questions into ChatGPT 3.5 and evaluated the response of it.

- (A) The accuracy of pulpitis.
- (B) The accuracy of TMJ disorder.
- (C) The accuracy of periodontal disease.
- (D) The accuracy of pericoronitis.
- (E) The accuracy of caries.
- (F) The accuracy of malocclusion.
- (G) The accuracy of oral cancer.

As a supplement, we evaluated recently released LLMs, including ChatGPT 4o, Llama 3.1–405b, Gemini 1.5 Pro, and Claude 3.5 Sonnet, from September 2–6, 2024 (Figures S6–S9). Table 2 summarizes the overall accuracy of these updated LLMs. Notably, only Claude 3.5 Sonnet demonstrated improved performance relative to its predecessor, exhibiting the ability of preliminary oral cancer diagnosis (Figure S9G).

Diagnostic accuracy with prompts

To investigate whether using specific prompts would influence the performance of ChatGPT 3.5, we explored its diagnostic capabilities when provided with prompts indicating a particular professional background. With prompt use, ChatGPT 3.5 provided 102 (58.3%) correct diagnoses and 37 (21.1%) partially correct diagnoses. Similar to the scenarios without prompts, ChatGPT 3.5 with prompting exhibited a complete inability to diagnose pulpitis (0/25, 0%) (Figure 3A) and preliminary diagnostic capabilities for TMJ disorders, periodontal disease, caries, and malocclusion (25/25, 100%) (Figures 3B–3F). ChatGPT 3.5 with prompting correctly diagnosed pericoronitis

2 times (8%) within the 5-day period and provided partially correct diagnoses 16 times (64%) (Figure 2D). Descriptions suggestive of oral cancer elicited conservative responses from ChatGPT 3.5 with prompting, with 24 (96%) partially correct diagnoses (Figure 3G). However, Wilcoxon signed-rank tests revealed no significant differences in diagnostic accuracy for pericoronitis ($p = 0.56$) or oral cancer ($p = 0.58$) between scenarios with and without prompting (Figures 3H and 3I). After using prompts, ChatGPT 3.5 still recommended seeking a dentist for all diseases (Tables S8–S14).

Differences in diagnostic accuracy between languages

To investigate whether using a non-English language for questioning would impact diagnostic accuracy, we translated English queries into Chinese and inputted them into ChatGPT 3.5 in a similar manner. In Chinese, ChatGPT 3.5 yielded 108 (61.7%) correct diagnoses and 38 (21.7%) partially correct diagnoses. Over a period of 5 days, ChatGPT 3.5 in Chinese was also able to provide preliminary diagnoses for TMJ disorders, periodontal disease, dental caries, and malocclusion (25/25, 100%)

Table 1. The overall diagnostic accuracy of multi LLMs

	ChatGPT 3.5	ChatGPT 4.0	PaLM 2	Claude 2	Llama 2	Copilot
correct diagnosis (%)	102 (58.3)	144 (82.3)	127 (72.6)	123 (70.3)	125 (71.4)	117 (66.9)
partially correct diagnosis (%)	35 (20)	9 (5.1)	17 (9.7)	27 (15.4)	1 (0.6)	8 (4.6)

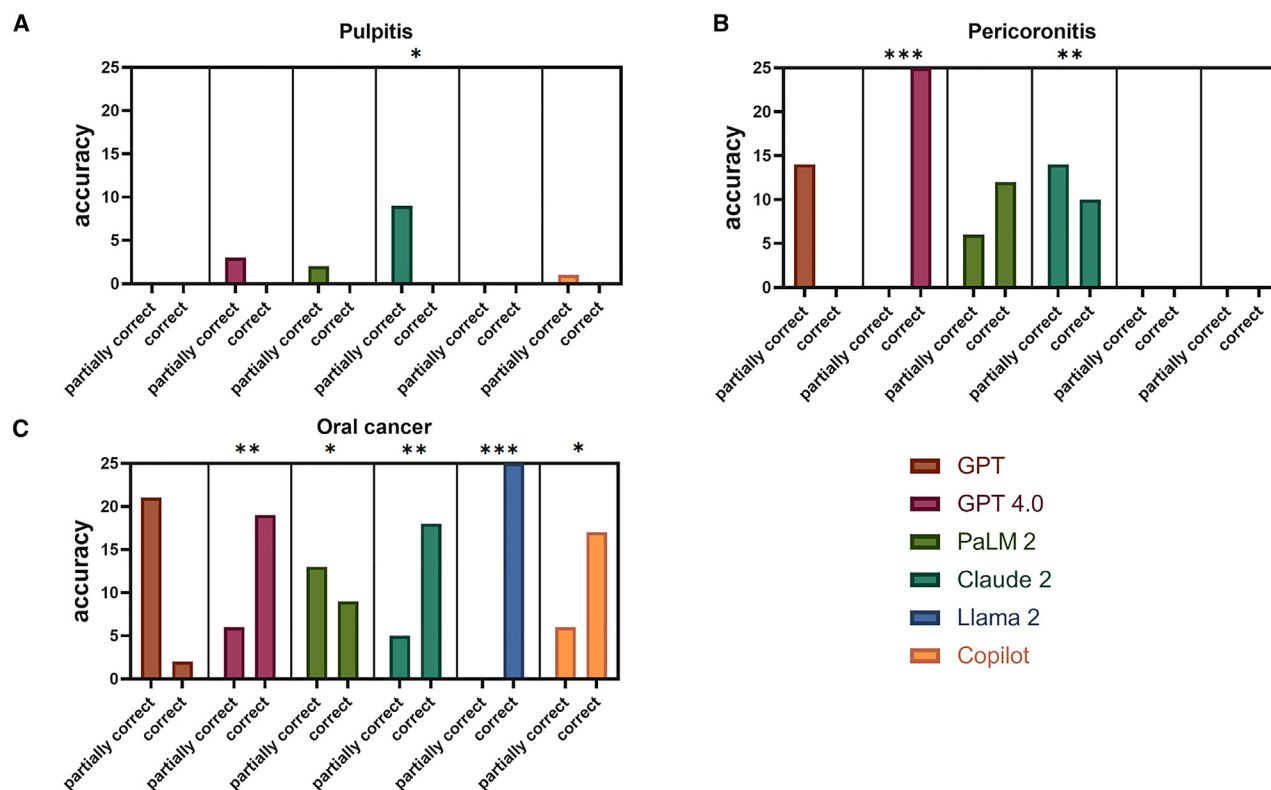


Figure 2. The accuracy comparisons of multiple LLMs in diagnosing common oral diseases

In a five-day experiment, five participants individually inputted formulated questions into various LLMs and evaluated the response of them. The details of specific LLMs' performance are shown at [Figures S1–S9](#). The diagnostic accuracy was compared with ChatGPT 3.5 in Friedman's rank test.

(A) The accuracy of pulpitis.

(B) The accuracy of pericoronitis.

(C) The accuracy of oral cancer.

∗: $p < 0.05$. ∗∗: $p < 0.01$. ∗∗∗: $p < 0.001$.

([Figures 4B–4F](#)). Surprisingly, ChatGPT 3.5 in Chinese provided a correct diagnosis for pulpitis 5 times (5/25, 20%) and a partially correct diagnosis 18 times (18/25, 72%) ([Figure 4A](#)). However, ChatGPT 3.5 in Chinese was completely unable to diagnose pericoronitis (0/25, 0%) ([Figure 4D](#)). For descriptions suggestive of oral cancer, ChatGPT 3.5 in Chinese provided 3 correct diagnoses (3/25, 12%) and 20 partially correct diagnoses (20/25, 80%). Wilcoxon signed-rank tests revealed a significant improvement in the diagnostic ability of ChatGPT 3.5 for pulpitis when questioned in Chinese ($p < 0.001$), while the diagnostic ability for pericoronitis significantly decreased ($p < 0.001$). There was no significant difference in the ability to judge suspected oral cancer ($p = 0.32$) ([Figures 4H–4J](#)).

When questioned in Chinese, ChatGPT 3.5 similarly recommended seeking medical attention with each inquiry. However, it is noteworthy that the recommendation statements used by ChatGPT 3.5 in Chinese were more monotonic, with “as soon as possible” being used 113 times (64.6%) ([Tables S15–S21](#)).

The technical report¹⁷ indicates that ChatGPT 4.0 has improved accuracy, especially in languages other than English, including Chinese, compared to ChatGPT 3.5. We further explore the accuracy of ChatGPT 4.0 by Chinese in diagnosing oral diseases. Generally, ChatGPT 4.0 made 141 (80.6%) times of correct diagnosis and 34 (19.4%) times of partially correct diagnosis. Same as other LLMs, ChatGPT 4.0 in Chinese could also provide preliminary diagnoses for TMJ disorders, periodontal disease, dental caries, and malocclusion (25/25, 100%) ([Figures 5B–5F](#)). In Chinese, the diagnostic ability of ChatGPT 4.0 for pulpitis and pericoronitis enhanced apparently ([Figures 5H and 5I](#), $p < 0.001$). For oral cancer, ChatGPT 4.0 in Chinese always give partially correct diagnosis (25/25, 100%), but did not make it its preferred diagnosis. The accuracy did not show significant difference compared to ChatGPT 3.5 ($p = 0.66$).

Table 2. The overall diagnostic accuracy of updated LLMs

	ChatGPT 4o	Llama 3.1–405b	Gemini 1.5 Pro	Claude 3.5 Sonnet
correct diagnosis (%)	100 (57.1)	100 (57.1)	103 (58.9)	125 (71.4)
partially correct diagnosis (%)	25 (14.3)	33 (18.9)	2 (1.1)	27 (15.4)

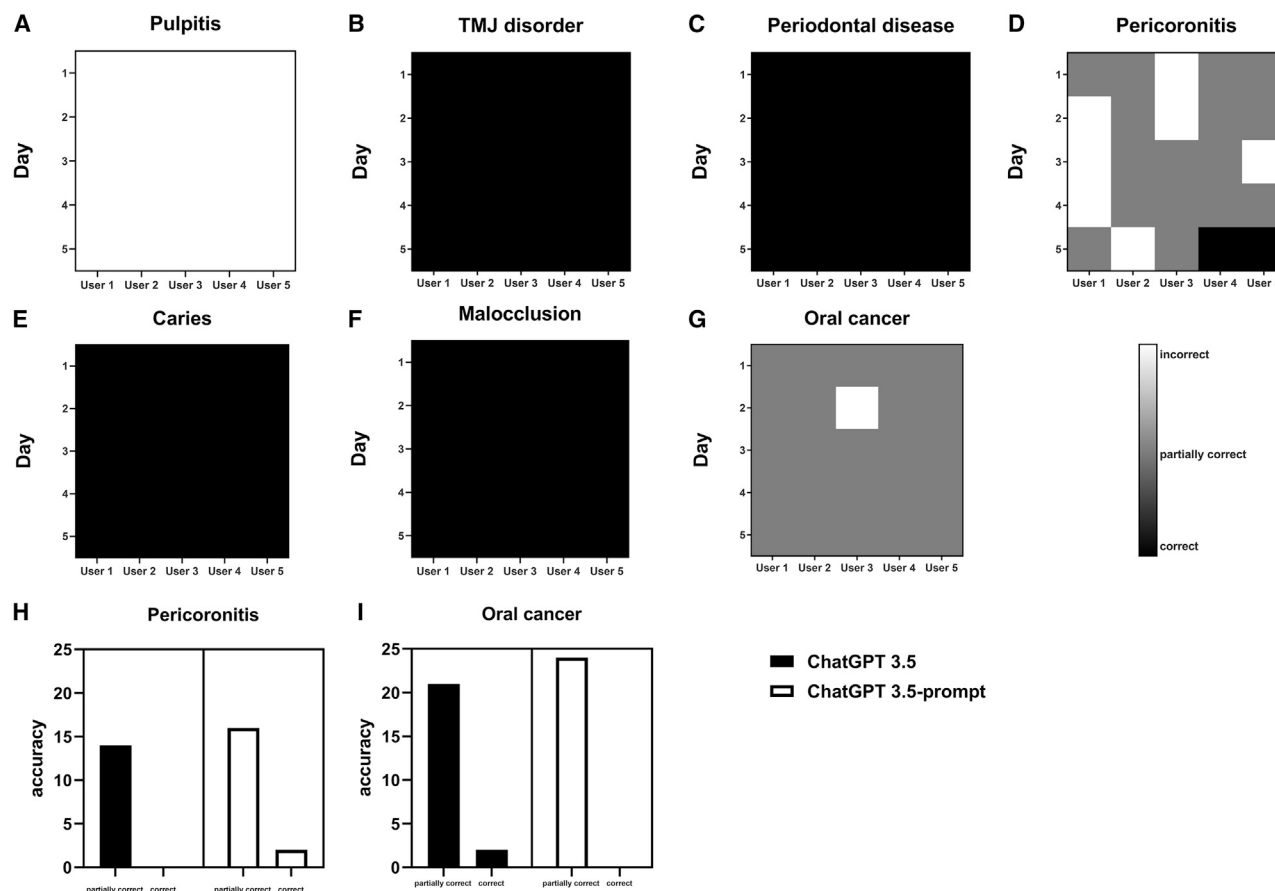


Figure 3. The accuracy of ChatGPT 3.5 with a background in diagnosing common oral diseases

In a five-day experiment, five participants individually inputted formulated questions into ChatGPT 3.5 after commanded it to play the role of a professional dentist and evaluated the response of it. The diagnostic accuracy was compared with original ChatGPT 3.5 in Wilcoxon signed rank test.

- (A) The accuracy of pulpitis.
- (B) The accuracy of TMJ disorder.
- (C) The accuracy of periodontal disease.
- (D) The accuracy of pericoronitis.
- (E) The accuracy of caries.
- (F) The accuracy of malocclusion.
- (G) The accuracy of oral cancer.
- (H) Comparison of diagnostic accuracy of pericoronitis before and after using prompt words.
- (I) Comparison of diagnostic accuracy of oral cancer before and after using prompt words.

DISCUSSION

As an exploratory, proof-of-concept study, we evaluated the potential utility of LLMs for self-diagnosis of common oral diseases. Over a 5-day period, five different experimenters posed specific questions to LLMs and assessed their provided diagnoses and medical advice. All evaluated LLMs demonstrated competence in preliminary diagnosis of TMJ disorders, periodontal disease, dental caries, and malocclusion. However, significant variability was observed in their ability to diagnose pulpitis, pericoronitis, and oral cancer. Notably, only ChatGPT 4.0 successfully identified pericoronitis, while Llama 2 was uniquely capable of diagnosing oral cancer. Additionally, all LLMs consistently suggested seeking medical attention in their

diagnoses, even when the original questions did not explicitly include relevant content.

However, the performance of multiple LLMs for diagnosing pulpitis, pericoronitis, and oral cancer varied. ChatGPT 3.5 showed a complete inability to diagnose pulpitis (0/25, 0%), and the performance of Claude 2, which exhibited better diagnostic ability, yielded only 9 (36%) partially correct diagnoses. The ChatGPT 4.0 only provided 3 (12%) partially correct diagnoses, this situation did not improve in ChatGPT 4o (0%, Figure S6A). This result may be attributed to the clinical symptoms of pulpitis, which can be similar to those of other conditions (including caries, dental abscess, and trigeminal neuralgia), leading to potential confusion. For pericoronitis, only ChatGPT 4.0 showed the ability to provide preliminary diagnoses (25/25, 100%), while other LLMs were

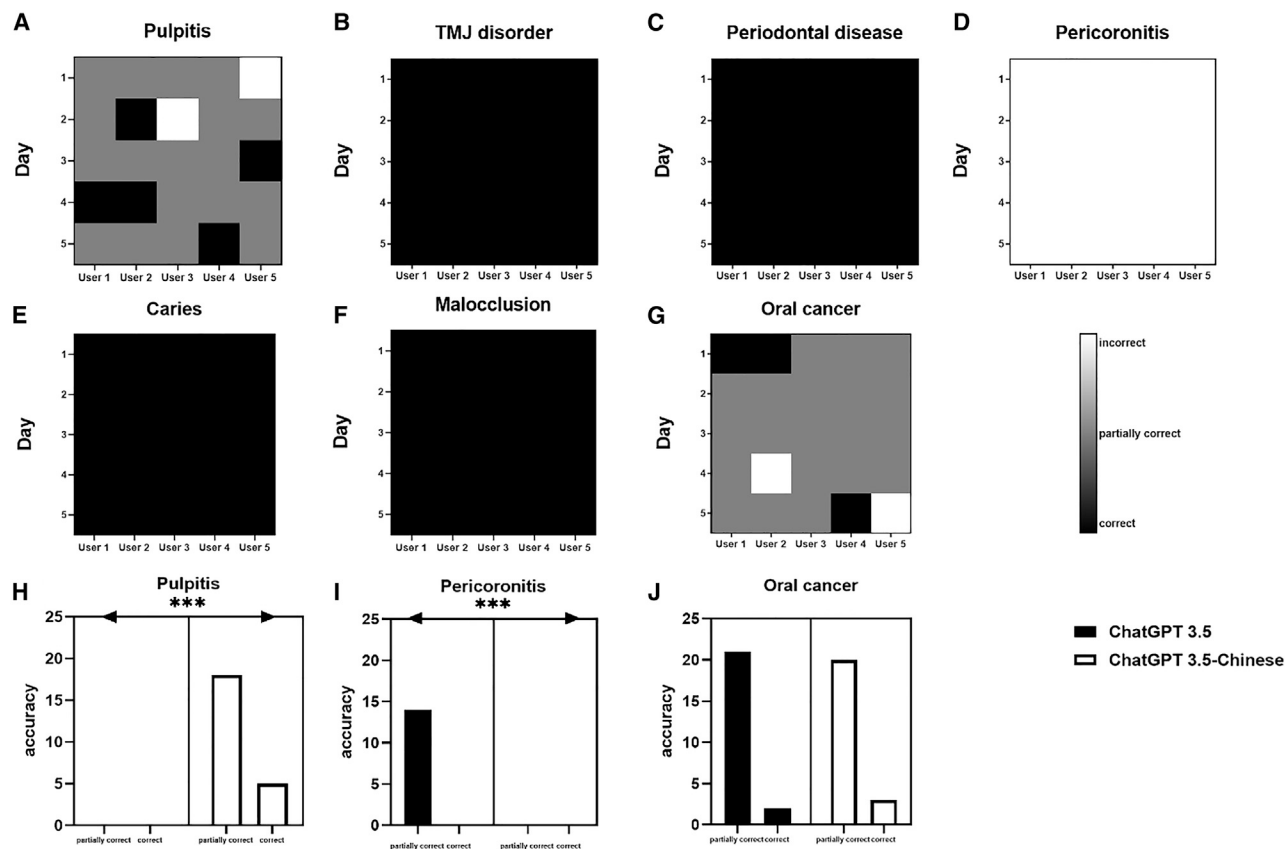


Figure 4. The accuracy of ChatGPT 3.5 by Chinese in diagnosing common oral diseases

In a five-day experiment, five participants individually inputted formulated Chinese questions into ChatGPT 3.5 after commanded it to play the role of a professional dentist and evaluated the response of it. The diagnostic accuracy was compared with original ChatGPT 3.5 in Wilcoxon signed rank test.

- (A) The accuracy of pulpitis.
- (B) The accuracy of TMJ disorder.
- (C) The accuracy of periodontal disease.
- (D) The accuracy of pericoronitis.
- (E) The accuracy of caries.
- (F) The accuracy of malocclusion.
- (G) The accuracy of oral cancer.
- (H) Comparison of diagnostic accuracy of pulpitis by English and Chinese.
- (I) Comparison of diagnostic accuracy of pericoronitis between English and Chinese.
- (J) Comparison of diagnostic accuracy of oral cancer by English and Chinese.

***: $p < 0.001$.

unable to make correct diagnoses. Similarly, the clinical manifestations of pericoronitis are similar to those of periodontal disease, potentially resulting in misdiagnosis by LLMs. Although caries may be combined with or secondary to other oral diseases, their characteristic features, such as oral defects and color changes, are more easily detected by patients. However, toothache may be the main symptom of pulpitis, pericoronitis, or periodontal disease, and therefore these diseases cannot be easily distinguished by the public.

For oral cancer, LLMs were generally able to provide accurate or partially accurate diagnoses and provided stronger recommendations for seeking medical attention. However, it is noteworthy that, except for Llama 2 and Claude 3.5 Sonnet, other LLMs exhibited instances of underdiagnosis during the experi-

mental period, highlighting the potential for errors in medical decision-making by patients in real health care applications.

To assess recent advancements in generative AI, we evaluated several newly updated LLMs. Surprisingly, with the exception of Claude 3.5 Sonnet, which demonstrated competence in preliminary oral cancer diagnosis, these models showed no significant improvement in oral disease diagnostic capabilities. This outcome underscores the persistent lack of emphasis on oral health within the AI development sector and broader society.

To investigate the impact of using prompts on the diagnostic performance of ChatGPT 3.5 for oral diseases, we instructed ChatGPT 3.5 to generate information about and respond as a dentist. After prompt use, the diagnostic ability for TMJ disorders, periodontal disease, caries, and malocclusion was

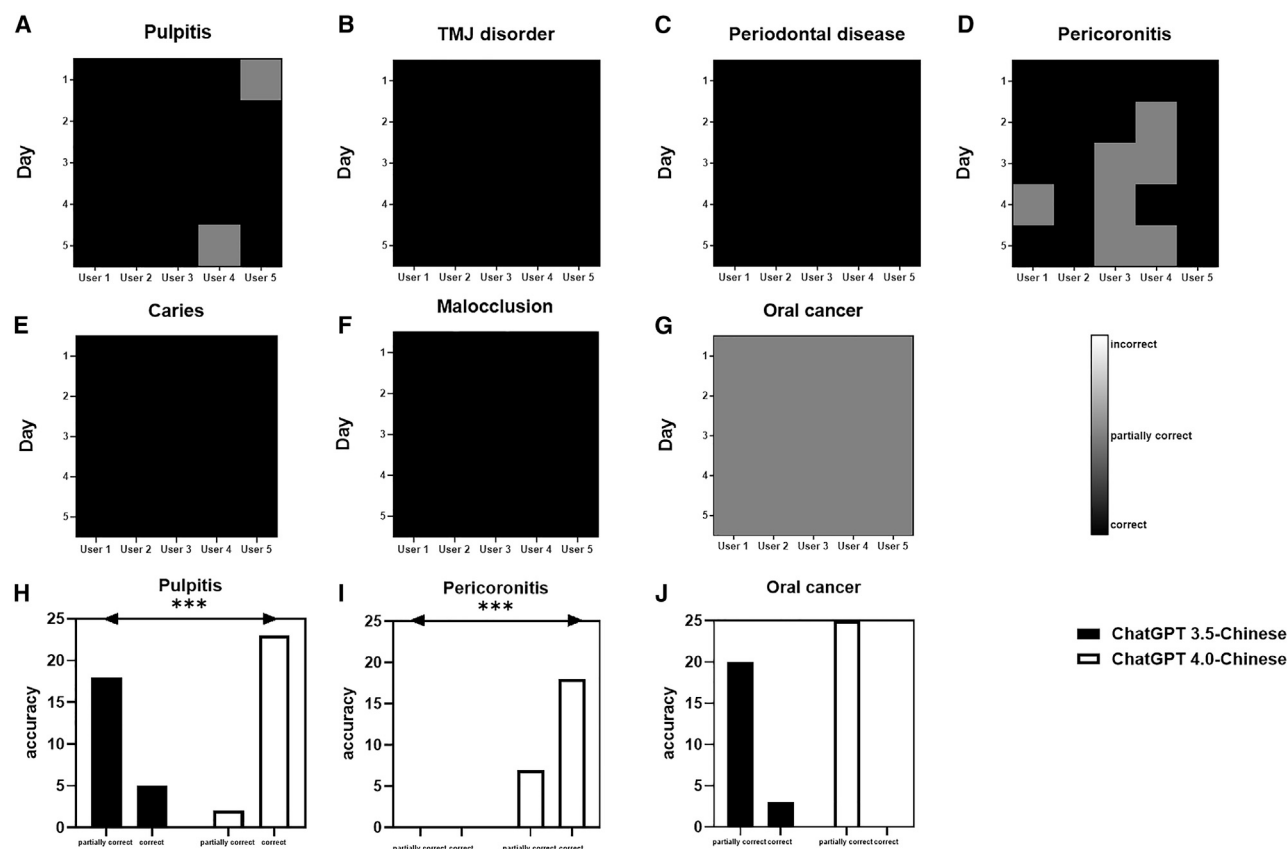


Figure 5. The accuracy of ChatGPT 4.0 by Chinese in diagnosing common oral diseases

In a five-day experiment, five participants individually inputted formulated Chinese questions into ChatGPT 4.0 after commanded it to play the role of a professional dentist and evaluated the response of it. The diagnostic accuracy was compared with original ChatGPT 3.5 in Wilcoxon signed rank test.

(A) The accuracy of pulpitis.

(B) The accuracy of TMJ disorder.

(C) The accuracy of periodontal disease.

(D) The accuracy of pericoronitis.

(E) The accuracy of caries.

(F) The accuracy of malocclusion.

(G) The accuracy of oral cancer.

(H) Comparison of diagnostic accuracy of pulpitis between ChatGPT 3.5 and 4.0.

(I) Comparison of diagnostic accuracy of pericoronitis between ChatGPT 3.5 and 4.0.

(J) Comparison of diagnostic accuracy of oral cancer between ChatGPT 3.5 and 4.0.

***: $p < 0.001$.

maintained (25/25, 100%). Using a prompt did not affect the diagnostic ability for pulpitis. For pericoronitis, ChatGPT 3.5 provided 2 correct responses (8%) and 16 partially correct responses (64%) when a prompt was used. In the case of oral cancer, ChatGPT 3.5 gave a partially correct response 24 times (96%). However, these differences were not significant. There are various methods for enhancing the performance of ChatGPT 3.5 in specific domains, and future research should involve more systematic evaluations to assess its potential applications in the medical field.

To assess the performance of ChatGPT 3.5 in a non-English environment, we conducted a pilot experiment to investigate the impact of using Chinese on the diagnostic accuracy of ChatGPT 3.5. Similar to its performance in English, ChatGPT

3.5 in Chinese exhibited the ability to diagnose TMJ disorders, periodontal disease, caries, and malocclusion. The diagnostic accuracy of ChatGPT 3.5 for oral cancer showed no significant difference when questioned in Chinese than when questioned in English ($p = 0.32$). However, ChatGPT 3.5 in Chinese exhibited a complete inability to diagnose pericoronitis (0/25, 0%), which was a significant deviation from its performance in English ($p < 0.001$).

Surprisingly, when using Chinese, the diagnostic ability of ChatGPT 3.5 for pulpitis significantly increased ($p < 0.001$), providing 5 correct diagnoses (20%) and 18 partially correct diagnoses (72%). The technical report¹⁷ indicates that ChatGPT 4.0 has improved accuracy, especially in languages other than English, including Chinese, compared to ChatGPT 3.5. This

improvement has also been reflected in the medical field.^{18,19} We further explore the accuracy of ChatGPT 4.0 by Chinese in diagnosing oral diseases. Dramatically, in Chinese, the diagnostic ability of ChatGPT 4.0 for pulpitis and pericoronitis has been further enhanced ($p < 0.001$). Notably, ChatGPT 4.0 did not make any incorrect diagnosis in the experiment when using Chinese, even performed better than that in English. This difference could be attributed to variations in the training data used for English and Chinese. The prevalence of the diseases in each country may also affect the responses. Although we tried to reduce the differences caused by translation, LLMs could not handle them in the same way. In addition, LLMs may not be developed or may not work in the same manner as in an English environment. These results highlight the importance of considering language differences when applying LLMs to the self-diagnosis of oral diseases and suggest that caution should be exercised in generalizing LLMs for such applications across different languages. At this time, most studies on LLMs have been conducted in English, which has limited the promotion of LLMs. Studies focusing on the application of LLMs in other languages are valuable. The performance of ChatGPT 4.0 in Chinese proved that with the update of training data, the scalability and adaptability of LLMs will be very worth looking forward to in global health applications.

We observed that LLMs consistently emphasized the importance of seeking medical attention in each response. In the case of malocclusion, ChatGPT 3.5 tended to provide mild recommendations, while for oral cancer, ChatGPT 3.5 leaned toward using stronger language. Given the lack of oral health awareness in developing countries³ and among low-income populations, the application of ChatGPT 3.5 could enhance the motivation and utilization of oral health care.

Tomoyuki Kuroiwa et al.²⁰ explored the potential use of ChatGPT as a self-diagnostic tool for common orthopedic diseases. The study revealed varying diagnostic accuracies for different orthopedic conditions, such as carpal tunnel syndrome, which achieved 100% accuracy, while cervical spondylosis with myelopathy had an accuracy of only 4%. The reasons for these differences were similar to those in our study, indicating that the nonspecific symptoms of certain diseases can lead to misdiagnosis. Another study²¹ assessed the accuracy of ChatGPT 3.5, ChatGPT 4.0, and Google Bard in addressing common ocular symptom queries. Their findings revealed that ChatGPT 4.0 achieved the highest accuracy at 89.2%, which aligns closely with our observed accuracy of 82.3% for oral disease diagnosis. A multicenter study²² evaluated the ability of ChatGPT 3.5 to solve clinical problems in oral and maxillofacial surgery, showing an accuracy rate of 87.2% in responding to clinical scenarios.

Currently, most research assessing the application of ChatGPT in health care focuses on clinical perspectives, using clinical vignettes^{9,11,23,24} to formulate questions for evaluation. In contrast, we took a primary care perspective, formulating questions based on common symptoms to enhance the generalizability of the research results.

This study represents an attempt to investigate the diagnostic accuracy of LLMs in common oral diseases and explore their potential applications. Additionally, we have released the open-domain oral disease Q&A dataset of our study, which

can be utilized for the training of AI models. To verify the efficacy of the data collected for LLM training and to provide practical examples for interested researchers, we adjusted the Baichuan2-7B-Chat model using our dataset. The modified model demonstrated improvements in both in-domain and out-of-domain data performance, exhibiting excellent performance in human evaluations. The details of the preview model can be found in the [Data S1](#). We systematically analyzed variations in the application of different LLMs in medical settings across various operational and linguistic environments, contributing to a more universally applicable understanding of the study outcomes.

Traditional approaches to improving dental health delivery include expanding clinical infrastructure, mitigating oral disease risk factors, and enhancing public health education. LLMs offer a complementary strategy, potentially increasing oral health awareness through personalized advice and timely recommendations for professional dental consultation. While LLMs cannot substitute for human expertise in primary dental care, they represent a cost-effective and efficient adjunct to existing oral health-care systems.

While there are variations in the diagnostic accuracy of LLMs for common oral diseases across different environments, they notably recommend dental examination in all cases. This finding highlights the potential application of LLMs in primary oral health care. Prior to widespread implementation of large language models (LLMs) in routine healthcare, several critical issues warrant careful consideration. Potential contamination of training datasets may compromise the reliability of certain LLMs' outputs, as highlighted by recent research.²⁵ Moreover, a comprehensive evaluation framework should extend beyond diagnostic accuracy to encompass safety protocols, clinical applicability, and patient impact assessments. These multifaceted considerations are essential to ensure responsible and effective integration of LLMs into healthcare systems.²⁶ In the future, further training on LLMs specifically tailored to medical scenarios²⁷ will be essential to expand the scope of their applications. And we are looking forward to more large prospective cohort studies regarding LLMs in medicine.

Limitations of the study

Our study has several limitations. First, in formulating the questions, the study did not utilize queries proposed by actual patients but rather relied on questions compiled from primary symptoms. This may introduce potential disparities with real-world primary health care scenarios. Second, we only assessed seven common oral diseases; therefore, a comprehensive representation of the diagnostic accuracy for all oral conditions is lacking. Third, due to the limited repeatability observed in practical usage, we did not conduct an in-depth analysis of the medical advice text generated by LLMs.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Yunzhi Lin (linda_dentist@fjmu.edu.cn).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- All the data of this study for analysis is contained in the article and appendix.
- The Open Domain Oral Disease Q&A Dataset has been deposited at GitHub and are publicly available as of the date of publication. Accession numbers are listed in the [key resources table](#).
- The details of the preview model can be found in the [Data S1](#). Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

ACKNOWLEDGMENTS

This study was funded by Natural Science Foundation of Fujian Province (2023J05131), Joint Funds for the Innovation of Science and Technology, Fujian Province (2023Y9061), Joint Funds for the Innovation of Science and Technology, Fujian Province (2023Y9092), Talent Introduction Scientific Research Project of the First Affiliated Hospital of Fujian Medical University (2022-YJRC3914), and The First Affiliated Hospital of Fujian Medical University Excellent Talent Program (YXRQN-ZCF2023).

AUTHOR CONTRIBUTIONS

S.Z.: conceptualization, formal analysis, methodology, software, investigation, resources, data curation, writing – original draft, visualization. Y.Z.: conceptualization, formal analysis, investigation, data curation, writing – original draft, visualization. S.L.: software, investigation, resources, data curation. X.C.: formal analysis, methodology, software, data curation. Y.X.: formal analysis, methodology. H.L.: software, data curation. Y. L.: conceptualization, formal analysis. C.Z.: data curation, writing – review & editing, visualization, supervision, project administration, funding acquisition. Y. L.: data curation, writing – review & editing, visualization, supervision, project administration, funding acquisition.

DECLARATION OF INTERESTS

All authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.isci.2024.111495>.

Received: May 7, 2024

Revised: September 17, 2024

Accepted: November 26, 2024

Published: November 29, 2024

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
ChatGPT	OpenAI	chatgpt.com
Claude	Anthropic	claude.ai
Llama	Meta AI	llama.com
Gemini	Google AI	gemini.google.com
Copilot	Microsoft	copilot.microsoft.com
PaLM	Google AI	ai.google/discover/palm2
The Open Domain Oral Disease Q&A Dataset	GitHub	https://github.com/LinesHogan/Open-Domain-Oral-Disease-QA-Dataset

METHOD DETAILS

Study design

To prevent bias, in a five-day (Dec 12-16, 2023) experiment, five participants individually inputted formulated questions into LLMs and documented the responses, along with descriptions of the perceived necessity for seeking medical attention. All participants employed the Chrome browser (Google, USA). To simulate real-life self-diagnostic scenarios, the researchers posed questions immediately in a new chat.

Questions

We selected four dental diseases and three oral surgery diseases, including 1. pulpitis, 2. temporomandibular joint (TMJ) disorder, 3. periodontal disease, 4. pericoronitis (or impacted wisdom tooth), 5. caries, 6. malocclusion, and 7. oral cancer. These diseases were selected by four dentist(S.Z., S.L., X.C. and Y.L.) through voting because they are common in clinical dental care¹ or oral surgery²⁸ and because they represent conditions in a variety of orofacial positions. Two dentists summarized the main symptoms of these diseases, referring to primary symptoms provided by the National Institutes of Health,^{29,30} the Mayo Clinic official website,^{31–33} the Cleveland Clinic official website,^{34,35} and other reputable sources. These symptoms were then compiled into a standardized format (Textbox S1) for input into the LLMs.

Prompts

Previous studies³⁶ have observed that instructing LLMs to be distinguished experts can improve their ability in specific fields. To explore the effectiveness of this approach in the self-diagnosis of oral diseases, we initially used ChatGPT 3.5 to generate information that simulated that of a dental professional (Textbox S2). Subsequently, before each inquiry, this generated information was input into ChatGPT 3.5, prompting the model to assume the role of a dental professional before responding to the questions.

The impact of using Chinese language on diagnostic accuracy

To investigate the performance of ChatGPT 3.5 in a Chinese language environment, two native Chinese-speaking dentists translated the questions provided in Textbox 1 into Chinese. Then, we adjusted the Chinese version following the guidelines for the process of cross-cultural adaptation of self-report measures,³⁷ in short, with the program "translation-integration-back-translation-group, assessment-expert, consultation-predictive, test-forming preliminary tool". Finally, the Chinese questions were input into ChatGPT 3.5, and the responses were recorded. The diagnostic accuracy was calculated, and the differences in diagnostic accuracy for each disease were compared between English and Chinese via statistical analysis.

LLMs selection

In the assessment of diagnostic accuracy and the determination of differences when using prompts or Chinese, ChatGPT 3.5 was employed as a representative of LLMs. Developed by OpenAI, ChatGPT 3.5 is widely utilized globally and is available free of charge; it has been adopted by most dental researchers.¹⁰ To evaluate differences in diagnostic models, ChatGPT 4.0 (OpenAI, USA), Copilot (Microsoft, USA), Claude 2-100k (Anthropic, USA), PaLM 2 (Google, USA), and Llama 2-70b (Meta, USA) were selected to assess variations in diagnostic accuracy between the models.

QUANTIFICATION AND STATISTICAL ANALYSIS

Accuracy evaluation

When evaluating the LLMs' responses, if the sole diagnosis or the first diagnosis provided was accurate, the response was categorized as "correct". If a secondary diagnosis offered by the LLMs was correct, the response was categorized as "partially correct". Conversely, if the correct diagnosis was absent, the response was deemed "incorrect". Two researchers (Shi. Z. and Yuan. Z.) evaluated the diagnostic performance and recommendations individually and cross-checked them. The accuracy of "correct" or "partially correct" classifications was recorded and calculated separately. This study opted not to use the standardized assessment tools for LLMs in medicine since we mainly focus on comparing the diagnostic accuracy among different LLMs.

Statistical analysis

Statistical analysis was conducted using SPSS 28.0 (IBM, USA). A heatmap was generated to visually represent the diagnostic accuracy of LLMs across users and days. The Wilcoxon signed rank test was utilized to compare the diagnostic accuracy between two groups. Friedman's rank test was applied to compare diagnostic accuracy across multiple LLMs, with Bonferroni correction employed to calculate significance (p -value). A p -value less than 0.05 was considered to indicate a significant difference. The p -value with Bonferroni's correction was multiplied by 6 for reporting.